

mouselab examples

Table of Contents

Simple visualisations of the anatomy (average_template).....	1
Default.....	1
Change resolution.....	2
Change slice position.....	3
Change slice orientation.....	5
Change data.....	6
Caveat.....	8
Visualising new data in CCF space.....	10
Default.....	10
Changing resolution.....	10
Changing the slice to be visualised.....	11
Visualising multiple views.....	13
Changing the slice location.....	14
Adding supplementary views.....	15
Combining anatomical and volumetric data.....	17
Combining data at different resolutions.....	18
Combining layouts.....	20
A discussion on creating mixed plots with different masks.....	21
Creating animations using videofig(s) from nihelp.....	33

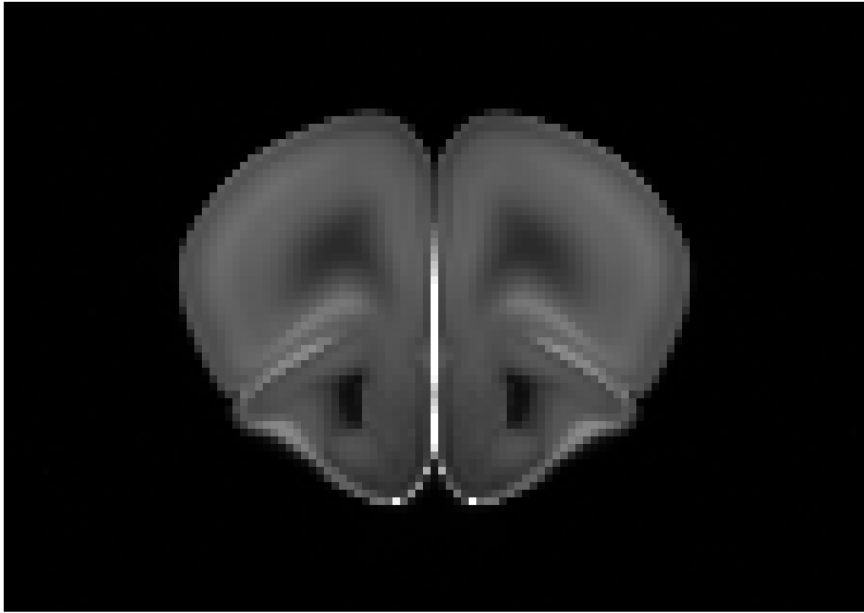
Simple visualisations of the anatomy (average_template)

Use the function `plotAllenSlice()` to plot a single plane of data from the Allen datasets in CCF space. They provide `average_template` (default), `ara_nissl`, and `annotation` data. By default, a coronal view is plotted at 100um resolution, 25% of the way along the volume.

Default

The first argument should be the type of data to be plotted ("`average_template`" (default), "`ara_nissl`", or "`annotation`").

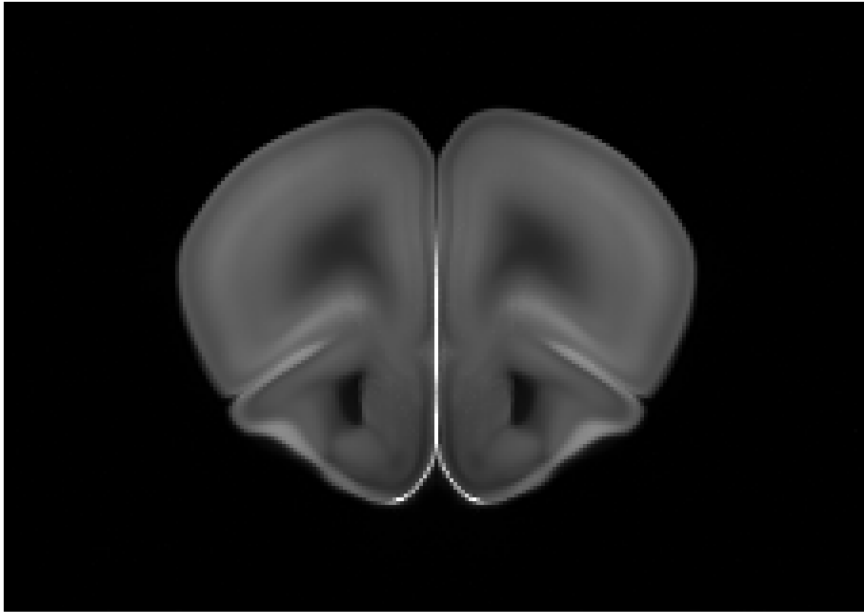
```
% Ignoring any inputs is the same as `plotAllenSlice("average_template")`  
figure; plotAllenSlice();
```



Change resolution

The default resolution is 100um. The repo also includes the data at 50um. The 10um and 25um can also be downloaded (from https://download.alleninstitute.org/informatics-archive/current-release/mouse_ccf/annotation/ccf_2017/) and added to the `annotation` directory.

```
% If specifying other parameters, the first argument must then be specified  
figure; plotAllenSlice('average_template', 'resolution', 50);
```

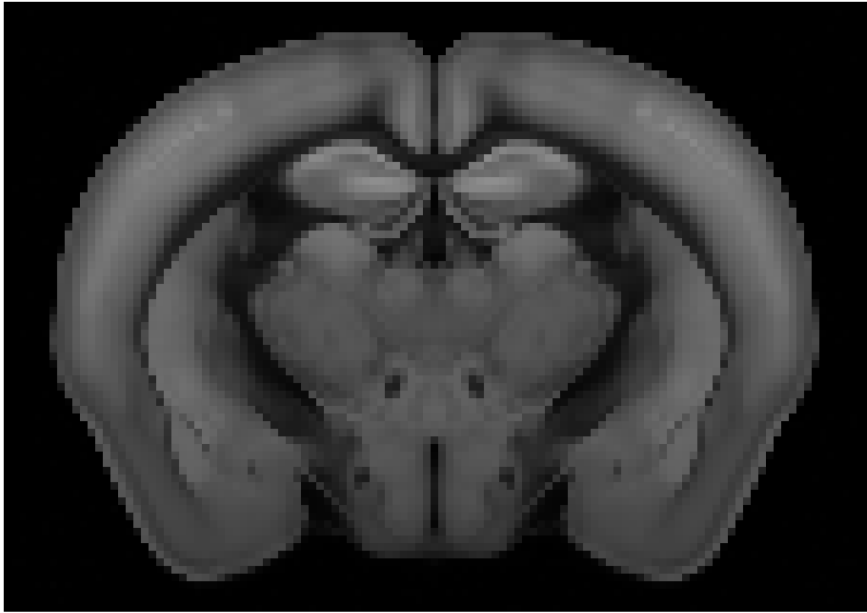


Change slice position

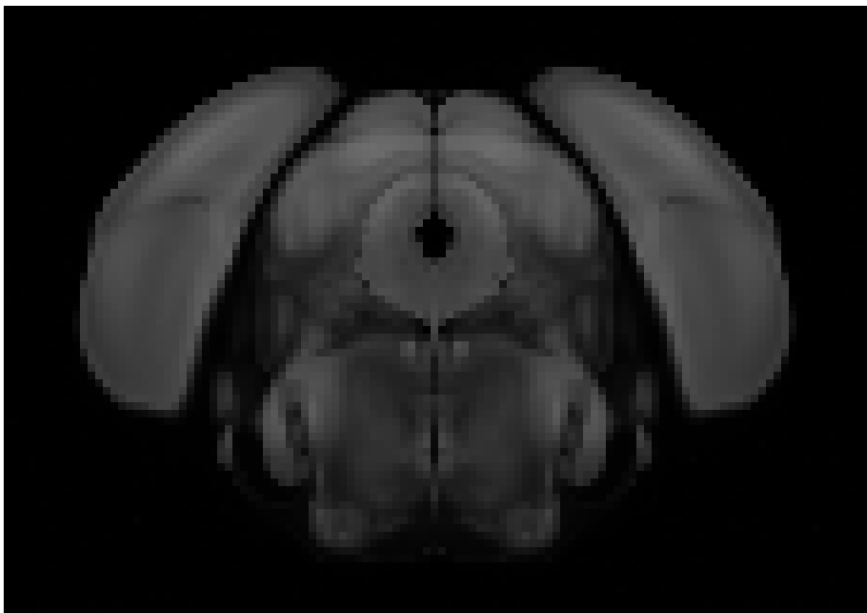
The slice position can be changed using `plotAllenSlice(___, 'slice', value)` syntax in 1 of 2 ways:

- If `value` is less than 1, the slice `value%` along that axis will be displayed. For example, if `value=0.5`, the middle slice will be displayed; if `value=0.25` (the default), the slice one-quarter of the way along that axis will be displayed.
- If `value` is greater than or equal to 1, the `value`-th slice will be displayed. For example, if `value=100`, the 100th slice will be displayed.

```
figure; plotAllenSlice('average_template', 'slice', 0.5);
```



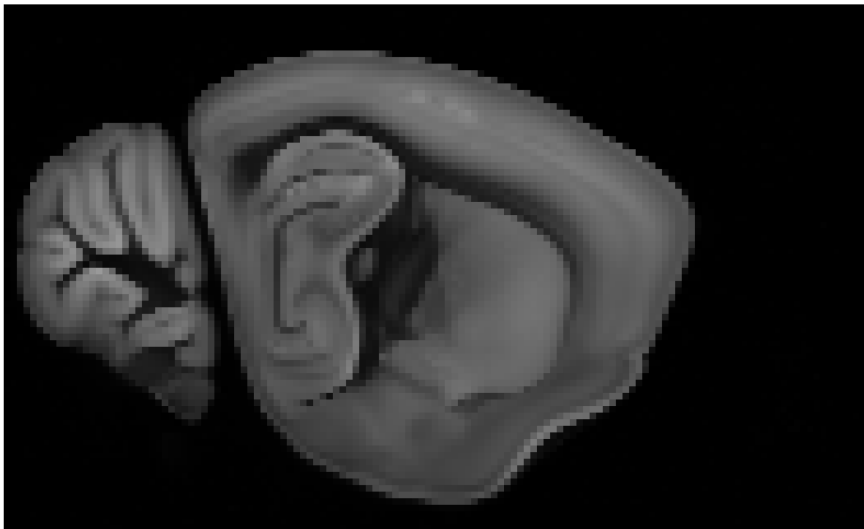
```
figure; plotAllenSlice('average_template', 'slice', 100);
```



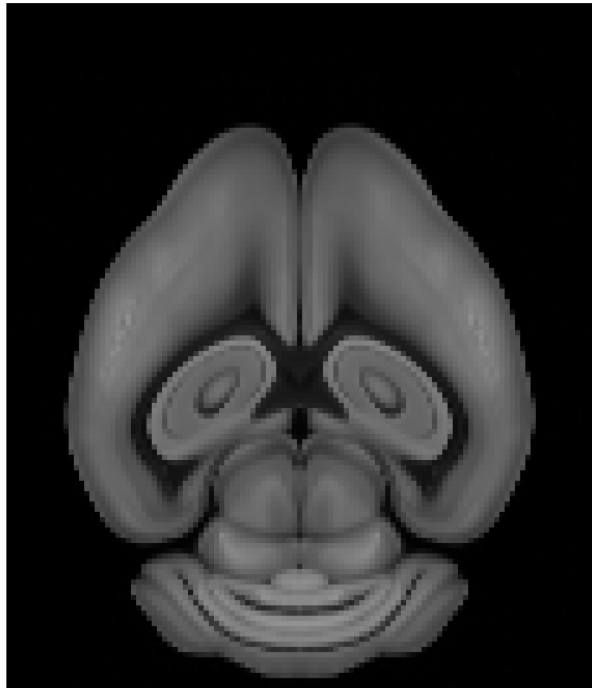
Change slice orientation

The axis along which to slice can be changed using `plotAllenSlice(___, 'dim', value)` syntax. Set `value` to 's' for sagittal sections, 't' for transverse sections, and 'c' for coronal sections (the default).

```
figure; plotAllenSlice('average_template', 'dim', 's');
```



```
figure; plotAllenSlice('average_template', 'dim', 't');
```



Change data

If desired, the data to plotted can be changed to the `ara_nissl` or `annotation` data from the Allen Institute.

```
figure; plotAllenSlice('ara_nissl', 'cmap', 'sky');
```



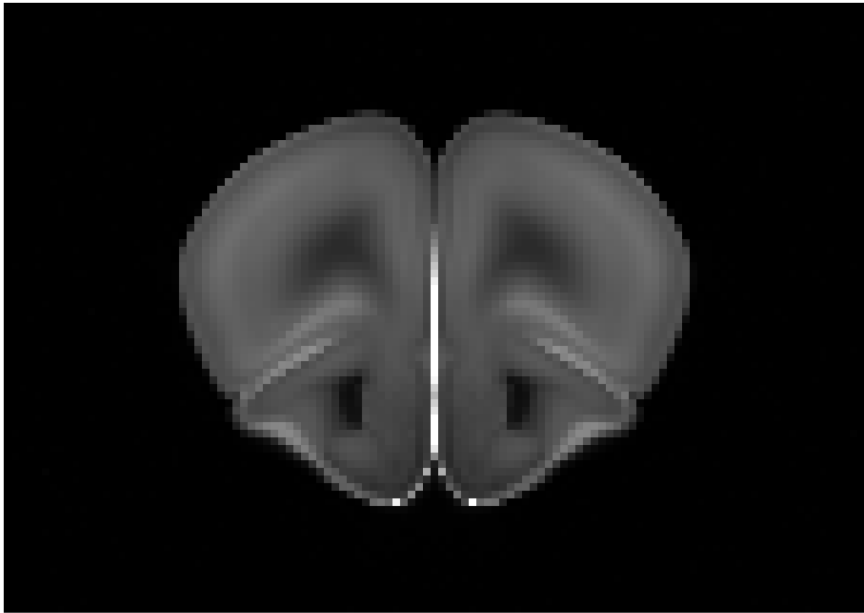
```
figure; plotAllenSlice('annotation', 'slice', 0.5, 'dim', 's');
```



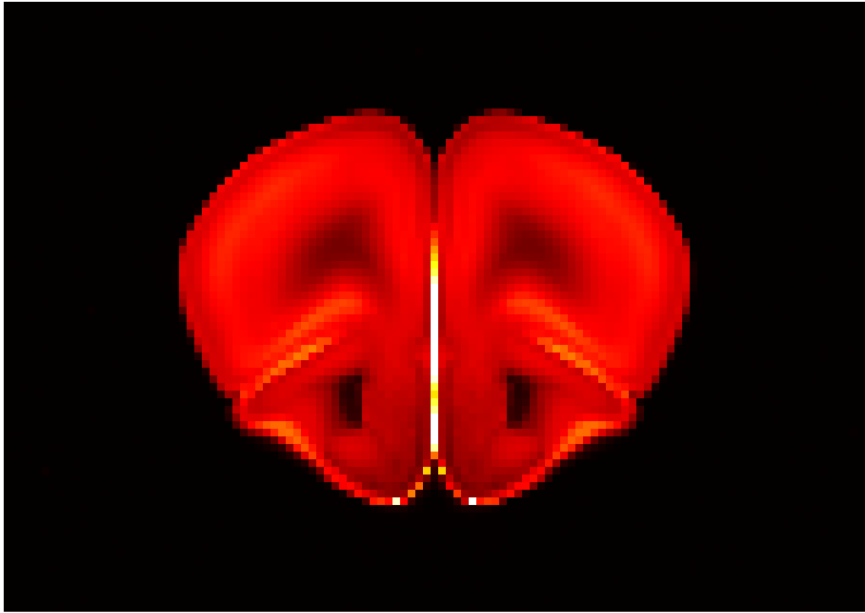
Caveat

The colors in the figure are 'true' colors i.e. the color is set at runtime (see [here/here/here](#) for more details). The color values cannot be changed after the figure is generated. However, they can be set using the `cmap` and `clims` name-value arguments. Note that these do not change the colormap/limits of the axes - they only change the way the data is generated for plotting.

```
figure; plotAllenSlice('average_template'); colormap(hot); % This will not have an effect
```



```
figure; plotAllenSlice('average_template', 'cmap', hot);
```



```
figure; plotAllenSlice('average_template', 'cmap', hot, 'clims', [1 300]);
```



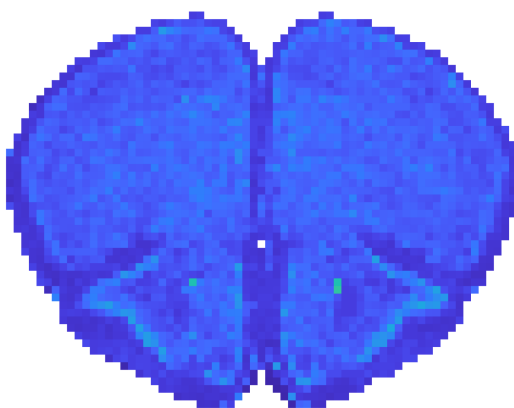
Visualising new data in CCF space

Here, we will use the `ara_nissl` data as example volumetric data in CCF space. But you can use your own data instead.

```
V = double(getAllenTemplate('ara_nissl')).*logical(getAllenTemplate('annotation'));
```

Default

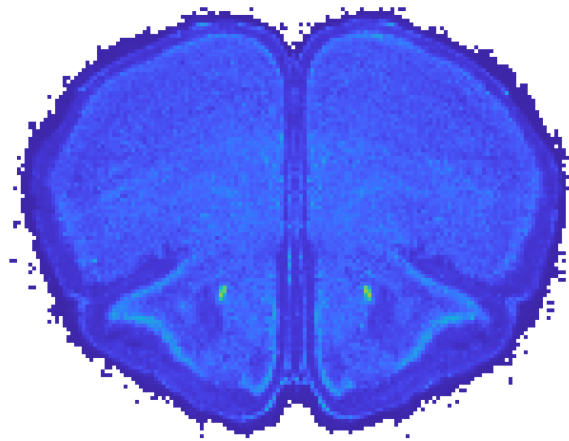
```
figure; plotCcfAnnotation(V);
```



Changing resolution

The `plotCcfAnnotation` function assumes your data is at 100um resolution and uses this to plot the data in 3D space. Let it know if you are plotting data at a different resolution.

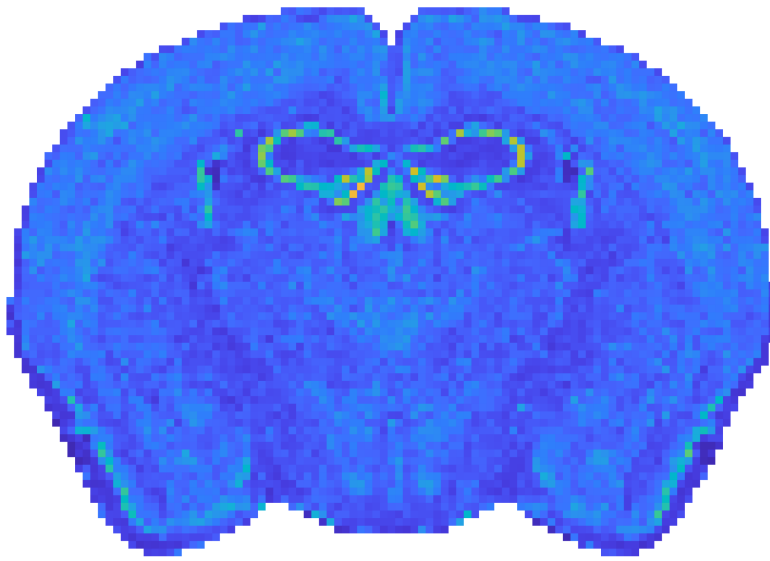
```
figure; plotCcfAnnotation(getAllenTemplate('ara_nissl', 50), 'resolution', 50);
```



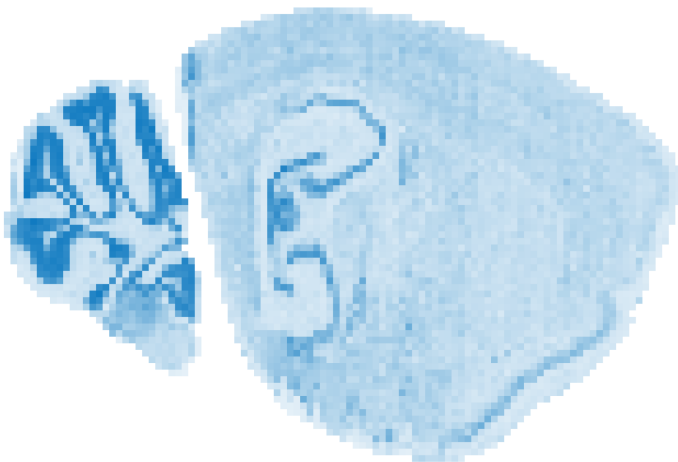
Changing the slice to be visualised

This can be changed the same as for the anatomy.

```
figure; plotCcfAnnotation(V, 'slice', 0.5);
```



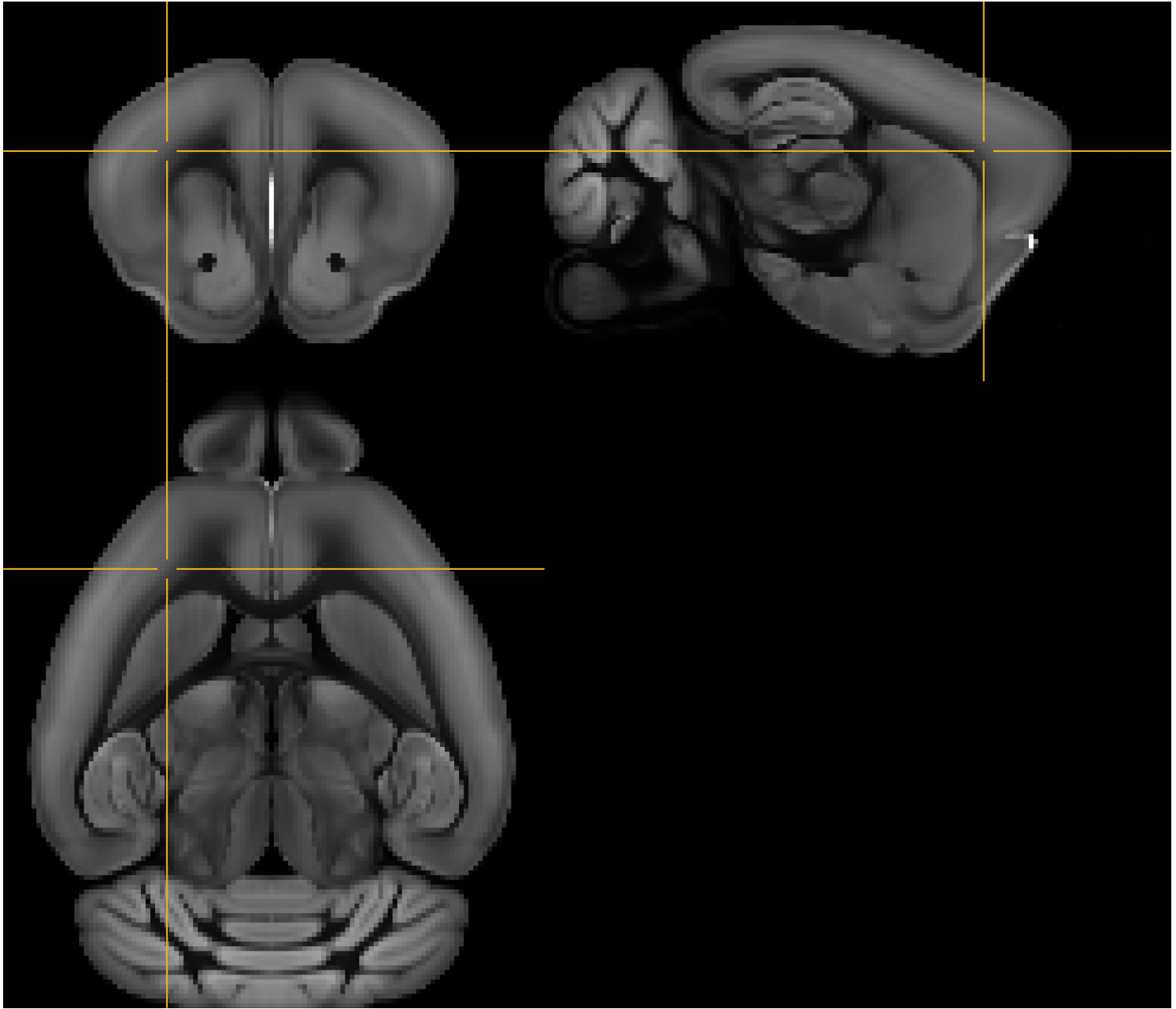
```
figure; plotCcfAnnotation(V, 'dim', 's'); colormap(sky);
```



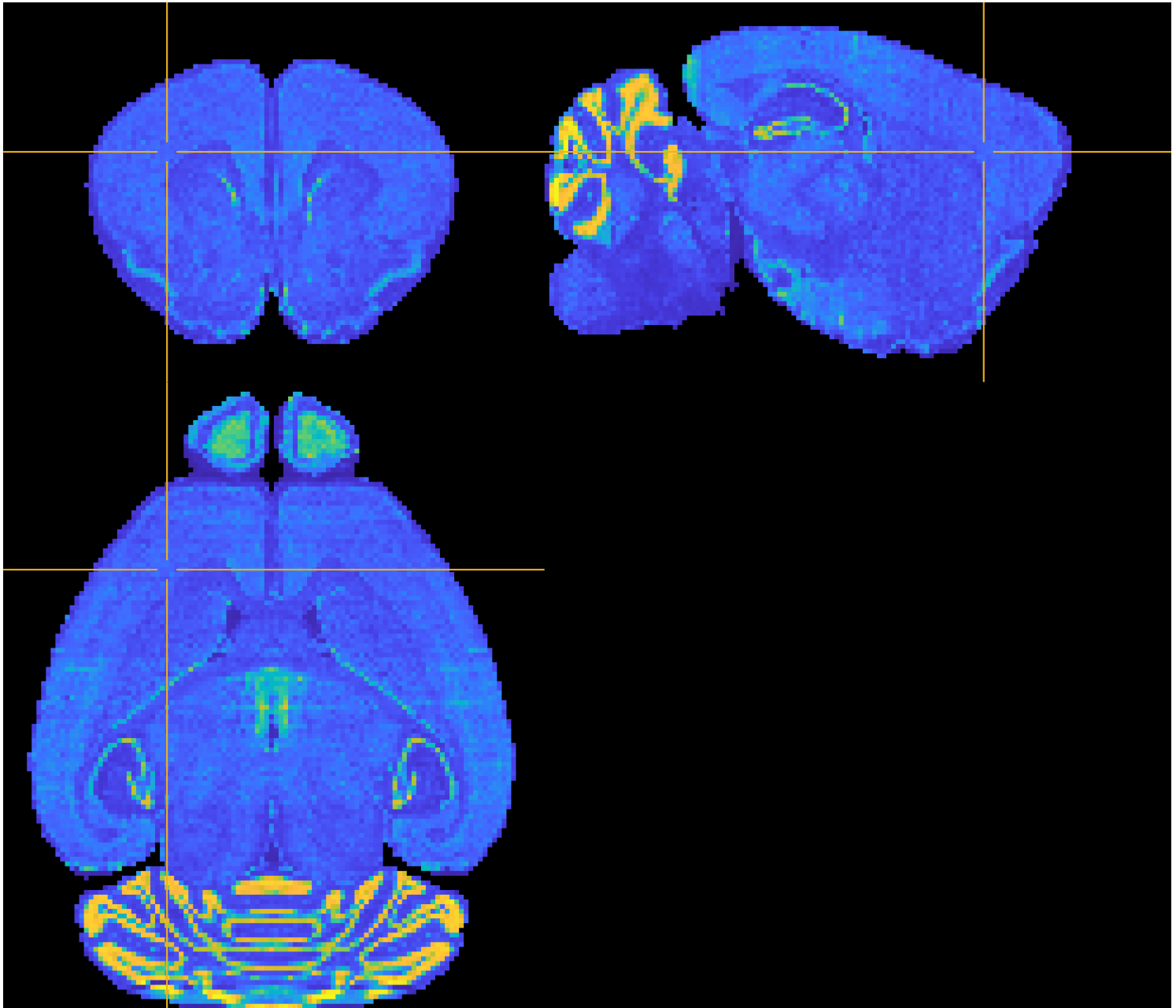
Visualising multiple views

Combined views in coronal, sagittal, and transverse planes can be generated using the functions `layoutAllenSlices` and `layoutCcfAnnotations`.

```
layoutAllenSlices();
```



```
V = double(getAllenTemplate('ara_nissl')).*logical(getAllenTemplate('annotation'));  
layoutCcfAnnotations(V);
```

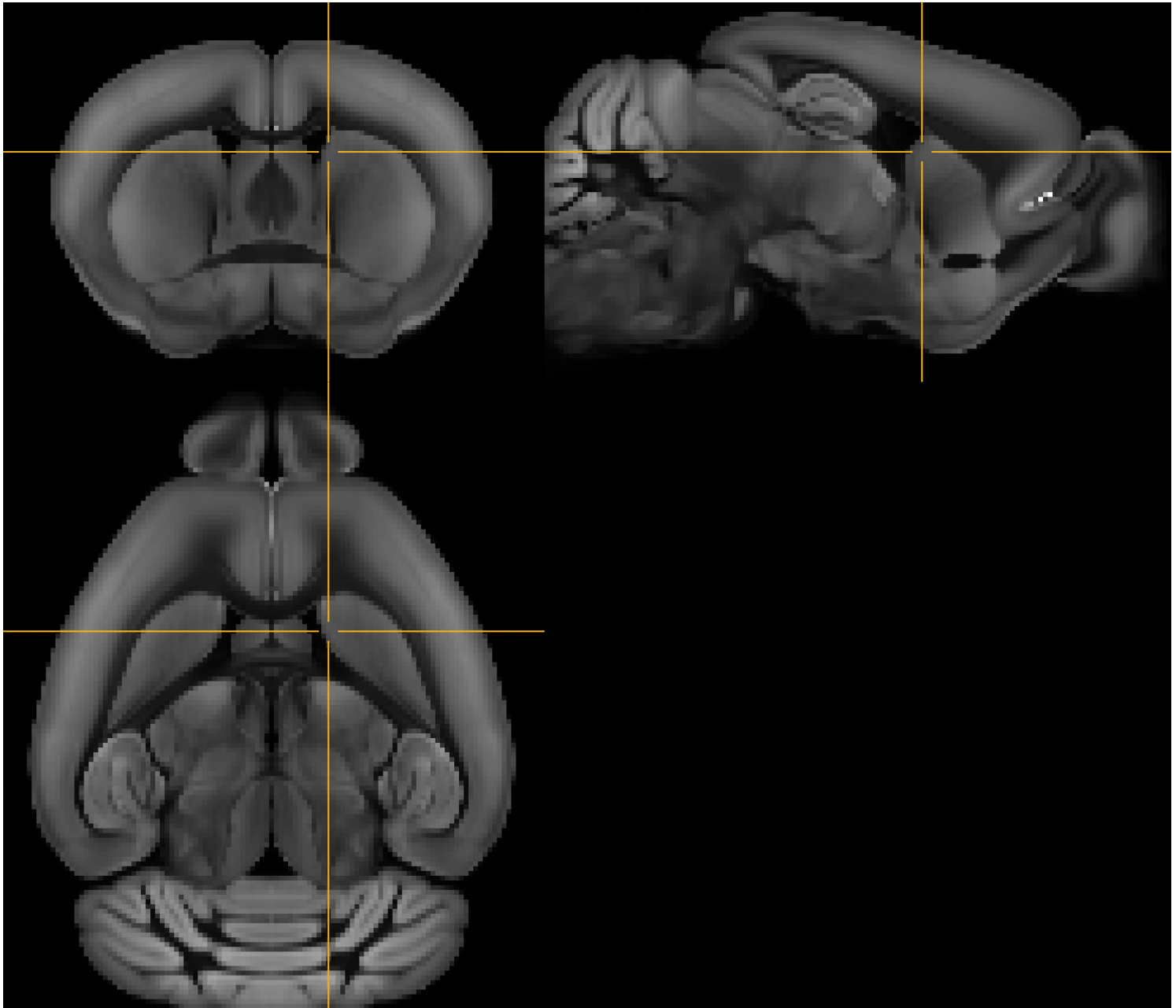


The crosshairs indicate where the slices in the other views have been made.

Changing the slice location

Use the 's', 'c', and 't' name-value arguments to change the position of each slice individually, using the syntax as described above.

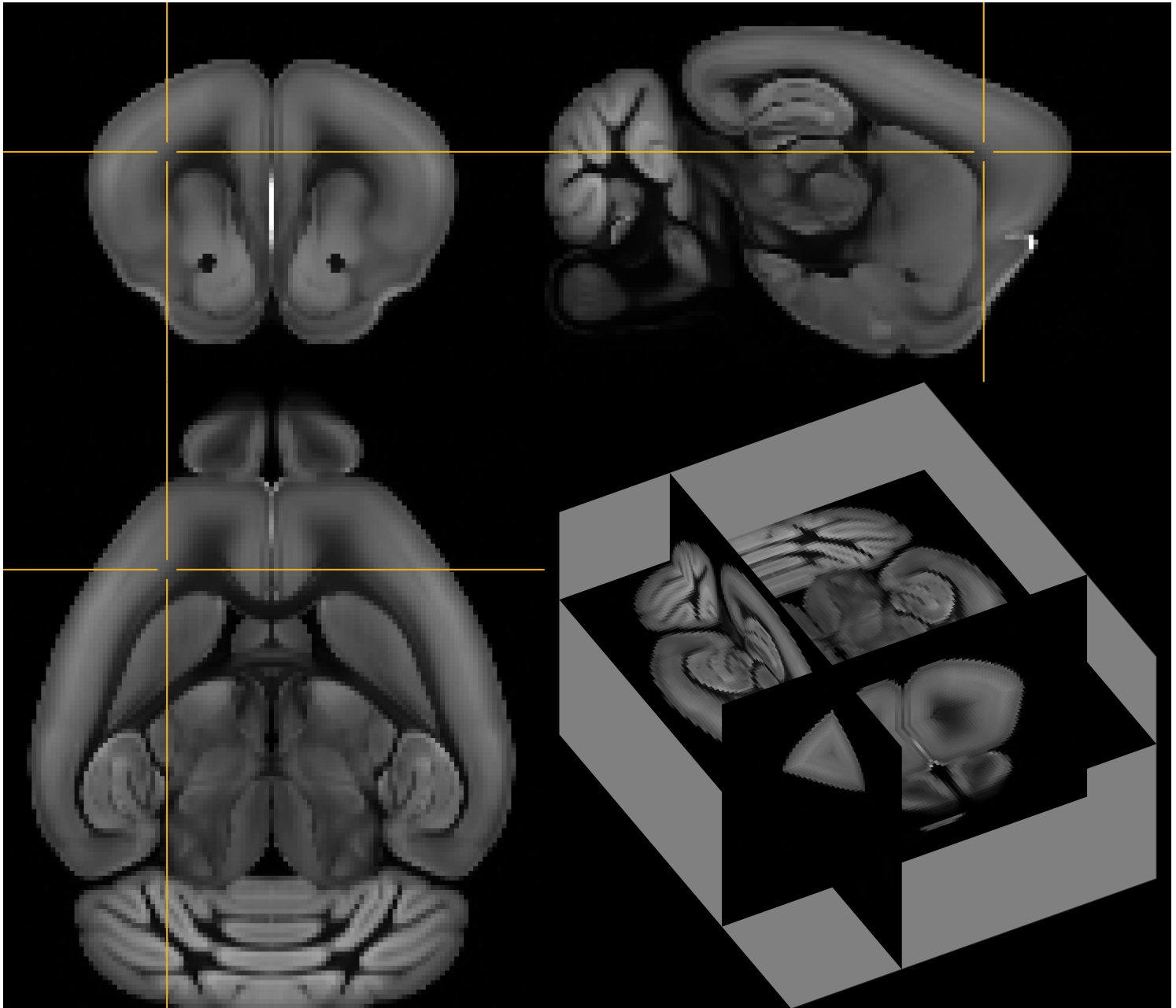
```
layoutAllenSlices('average_template', 's', .4, 'c', .4, 't', .4);
```



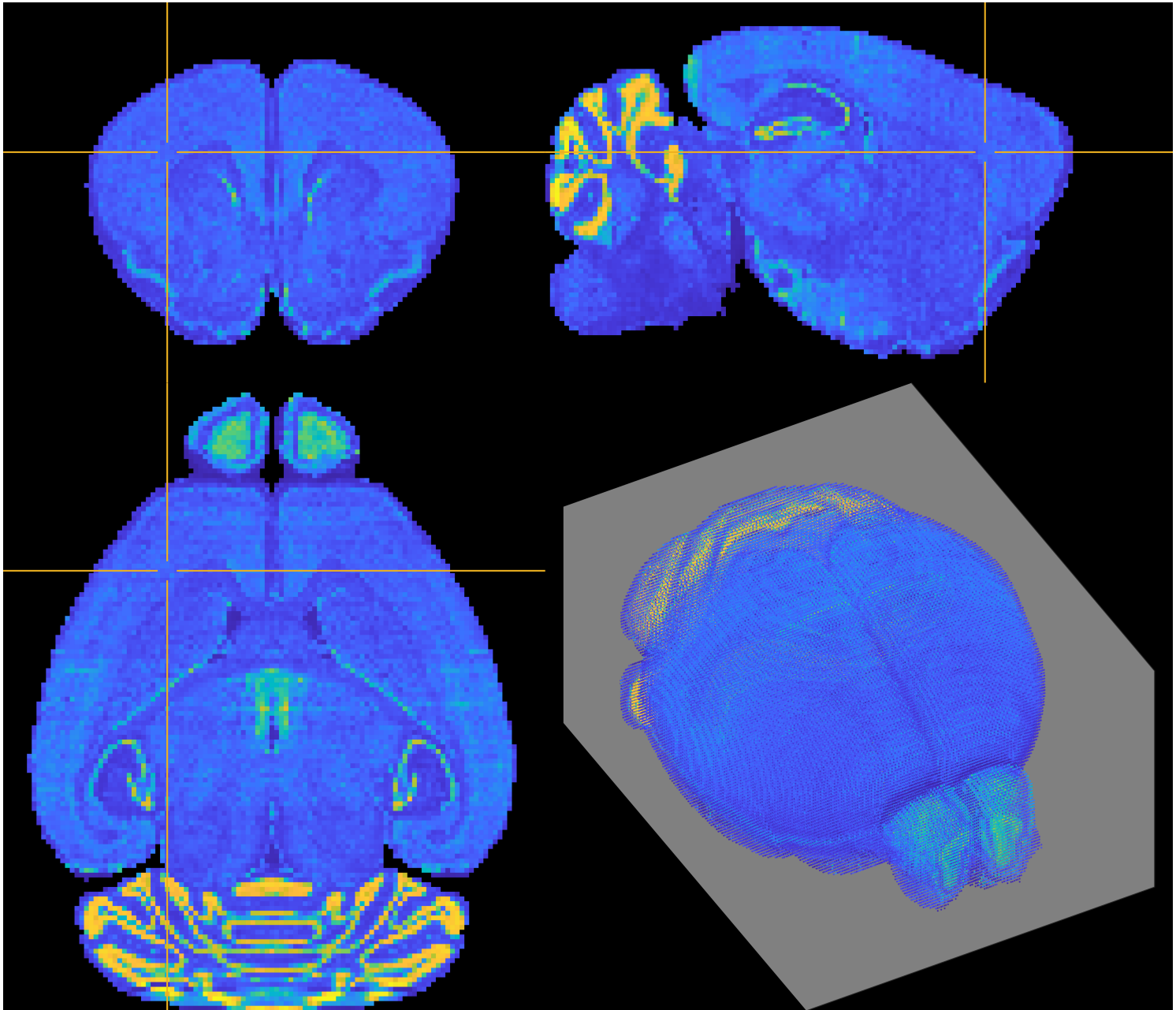
Adding supplementary views

Slices, surface, or volume data can be added in the remaining quadrant of the above image.

```
layoutAllenSlices('average_template', 'add', 'slices');
```



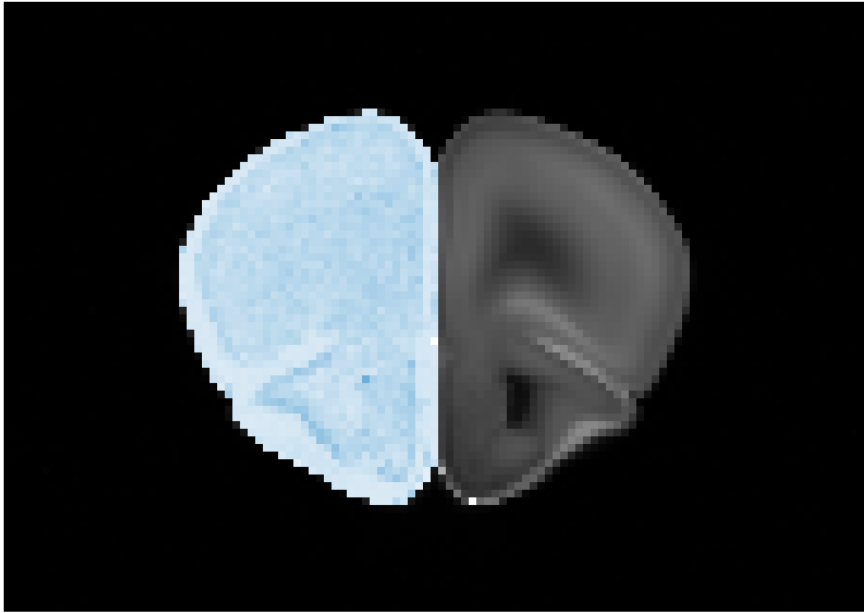
```
layoutCcfAnnotations(V, 'add', 'volume', 'vertices', V2v(V, getAllenTform(100), 1));
```



Combining anatomical and volumetric data

Volumetric and anatomical data can be plotted concurrently on a set of axes:

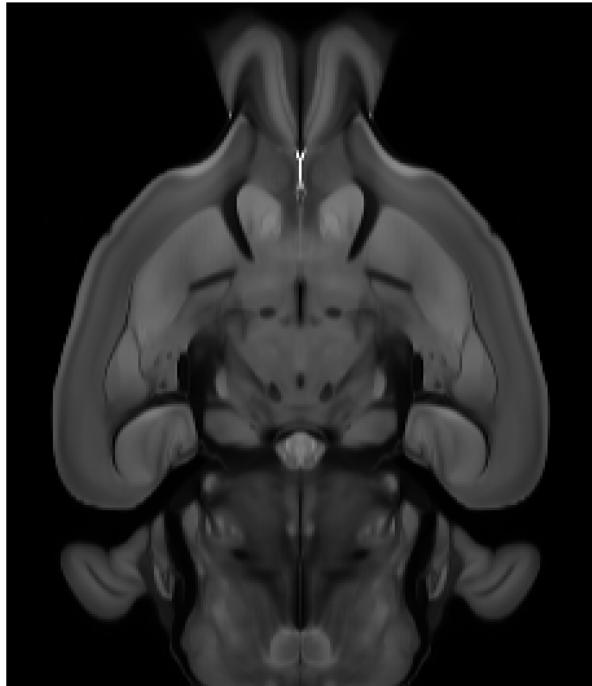
```
V = maskR(double(getAllenTemplate('ara_nissl')).*logical(getAllenTemplate('annotation')));
% example volumetric data
figure; plotAllenSlice();
hold on; plotCcfAnnotation(V);
colormap('sky'); % beacuse the volumetric data is scaled color and the anatomical data is
true color, the colormap can be change.
```



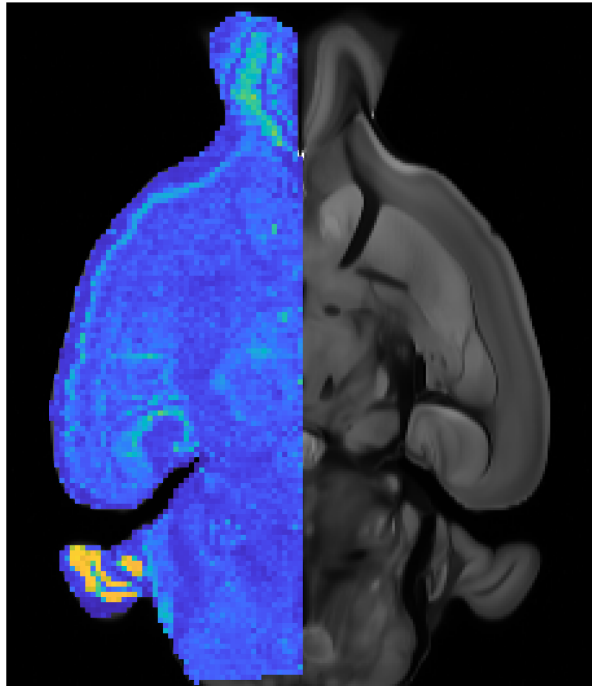
Combining data at different resolutions

Sometimes data at a higher resolution can mask data at a lower resolution. Use the `"(___ 'roundResolution', Y)"` name-value argument to plot data at a resolution X such that it is compatible with data at a resolution Y.

```
% This will not look good as the figures do not exactly line up
figure; plotAllenSlice('average_template', 'resolution', 50, 'slice', 0.7, 'dim', 't');
hold on; plotCcfAnnotation(V, 'slice', 0.7, 'dim', 't');
```



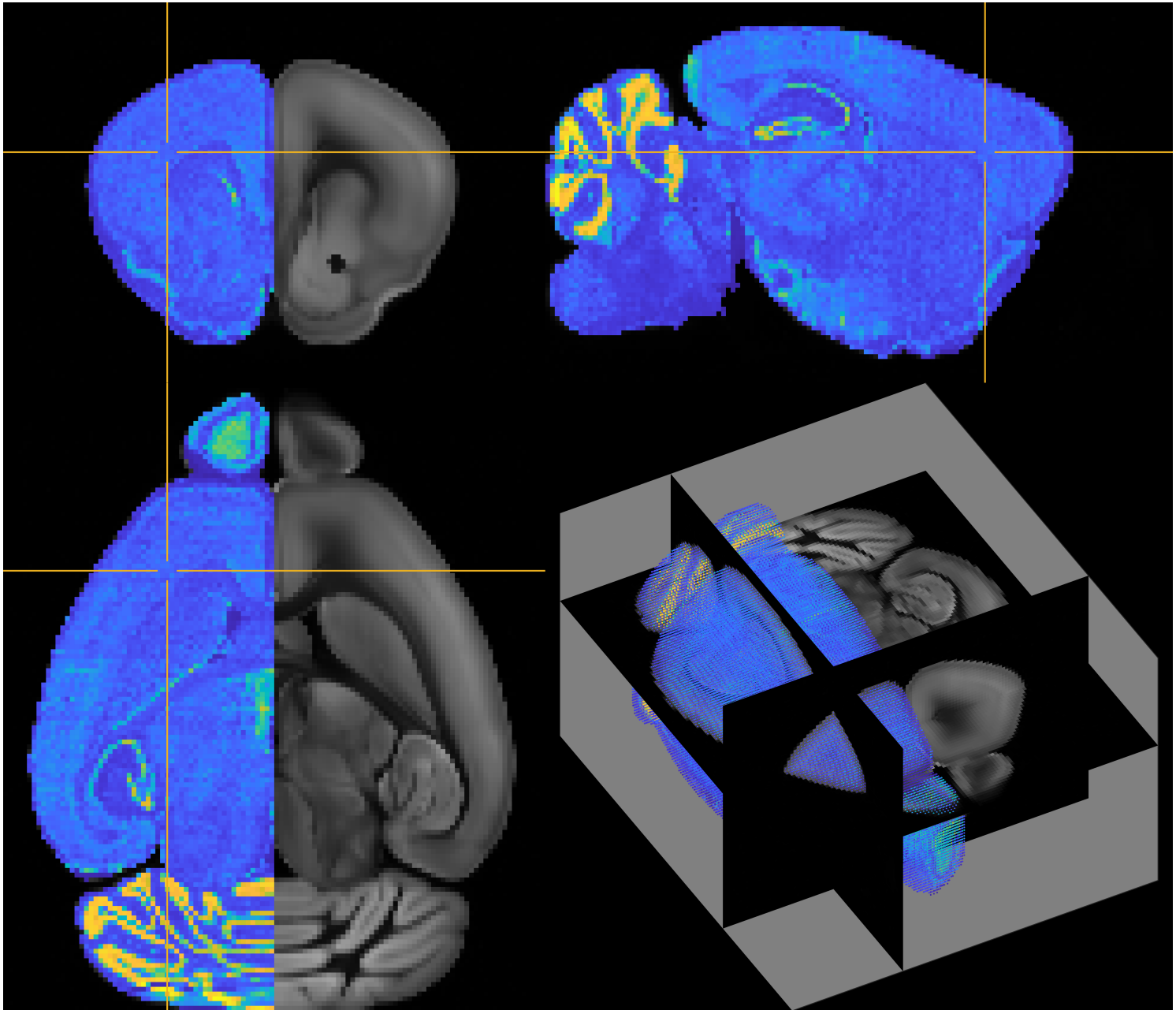
```
% Using `roundresolution` fixes it  
figure; plotAllenSlice('average_template', 'resolution', 50, 'slice', 0.7, 'dim', 't',  
    'roundResolution', 2);  
hold on; plotCcfAnnotation(V, 'slice', 0.7, 'dim', 't');
```



Combining layouts

Use the output from `layoutAllenSlices` with the 'Parent' name-value argument in `layoutCcfAnnotations` to plot multi-view data concurrently:

```
fig = layoutAllenSlices('add', 'slices');  
layoutCcfAnnotations(V, 'Parent', fig, 'add', 'volume');
```

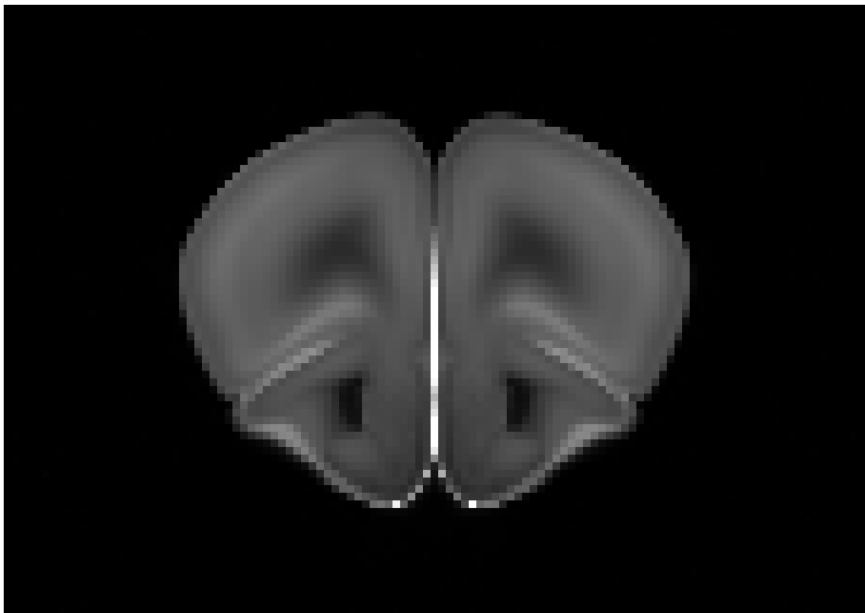
A discussion on creating mixed plots with different masks

The Allen data includes 'annotation' and 'average_template' data. The 'average_template' data is similar to T1 and can be used for plotting MR-style images. However, it includes a lot of background noise. The 'annotation' data contains a parcellation of the whole brain, and can further be binarised to be used as a mask.

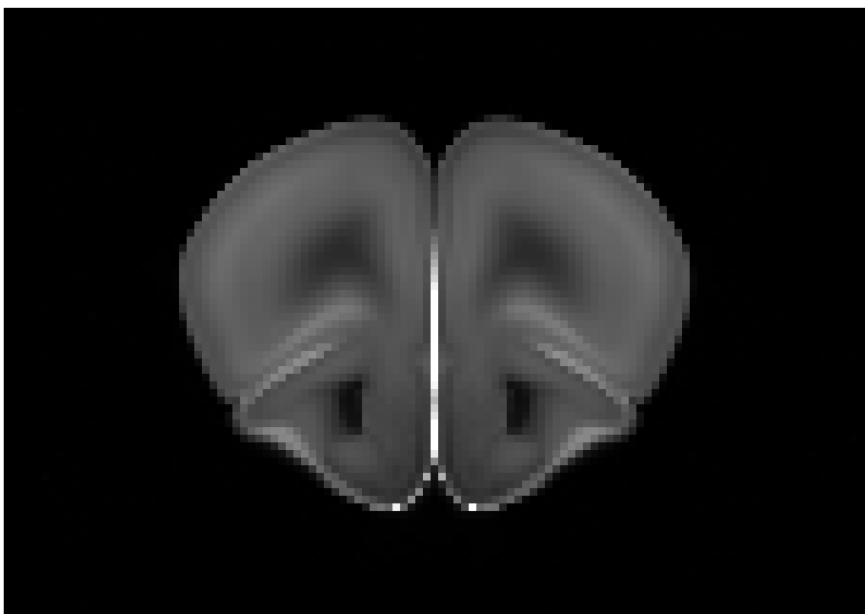
`plotAllenSlice` can plot this data. By default, it uses a gray colormap and plots the whole volume (zeros, nonzeros etc) i.e. no mask is applied.

```
% These are all the same (the supplied arguments are the defaults)
```

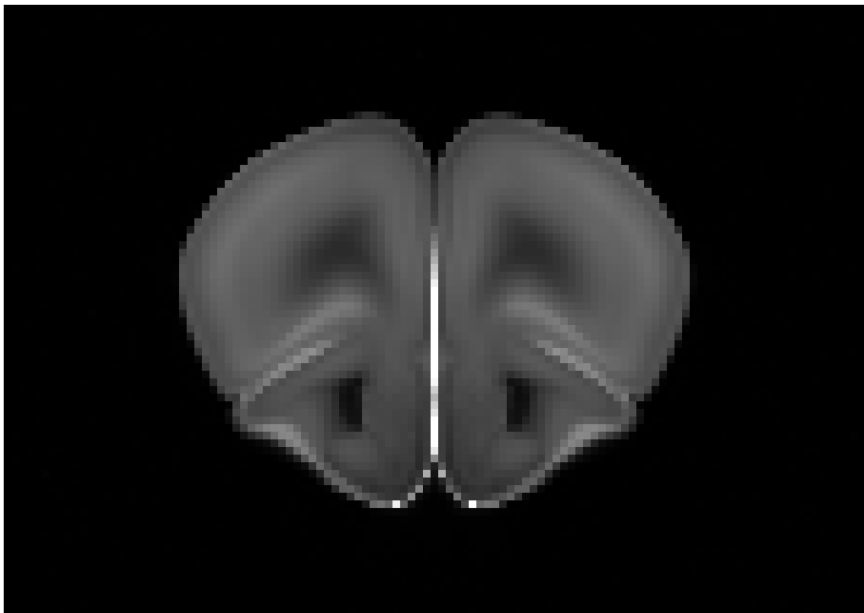
```
figure; plotAllenSlice();
```



```
figure; plotAllenSlice('average_template');
```

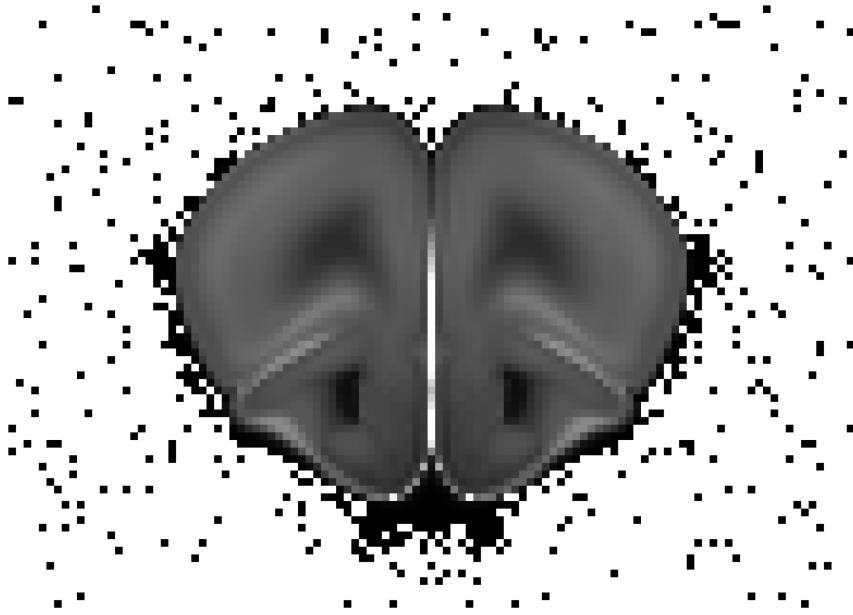


```
figure; plotAllenSlice('average_template', 'mask', @(x) true(size(x)));
```



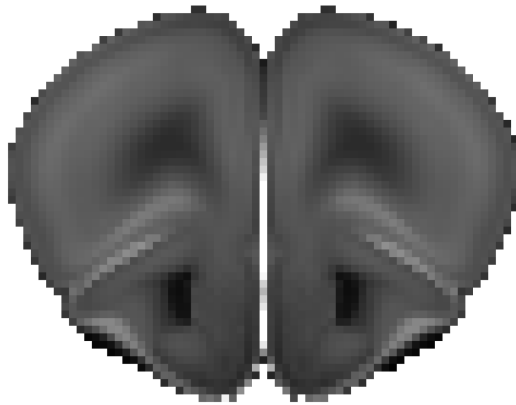
Note that some but not all of the background voxels are zero, and so applying a logical mask looks bad.

```
figure; plotAllenSlice('average_template', 'mask', @logical);
```



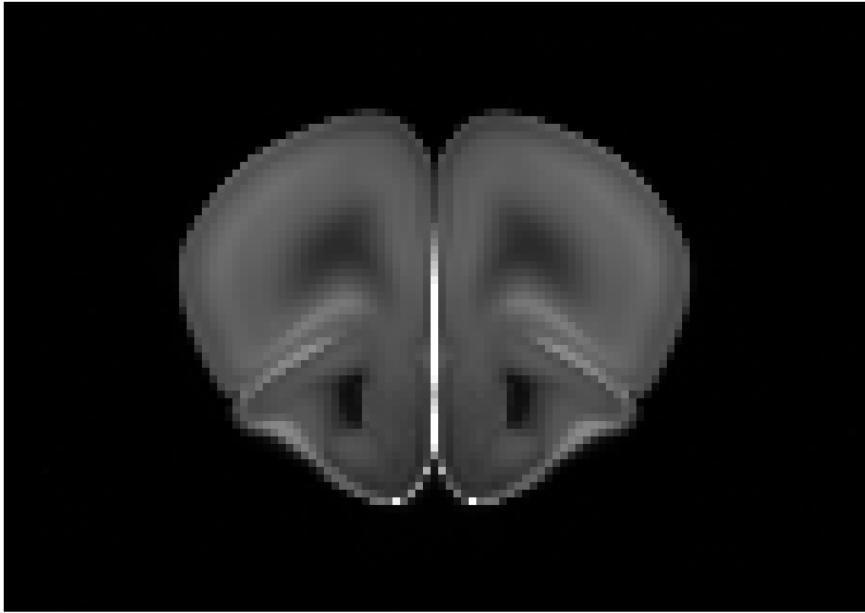
In contrast, using the 'annotation' as a mask results in only the brain voxels being plotted.

```
figure; plotAllenSlice('average_template', 'mask', getAllenTemplate('annotation'));
```

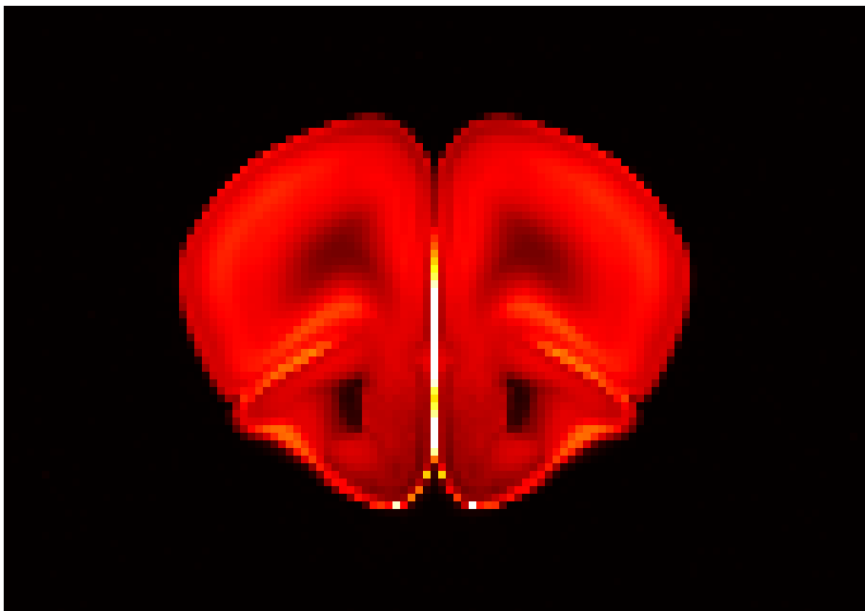


The colormap of `plotAllenSlice` cannot be changed after figure generation, as a true colour image is being plotted. However, it can be changed at runtime:

```
figure; plotAllenSlice('average_template'); colormap('hot'); % this won't work
```



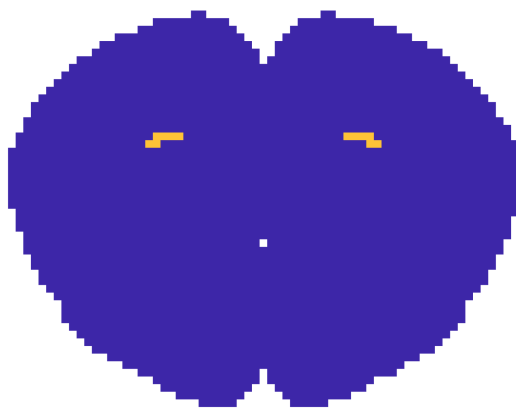
```
figure; plotAllenSlice('average_template', 'cmap', 'hot'); % this will
```



Similar results can be achieved using `plotCcfAnnotation`. This is because `plotAllenSlice` is a wrapper for `plotCcfAnnotation` that simply converts the Allen data to true colour volumes before plotting.

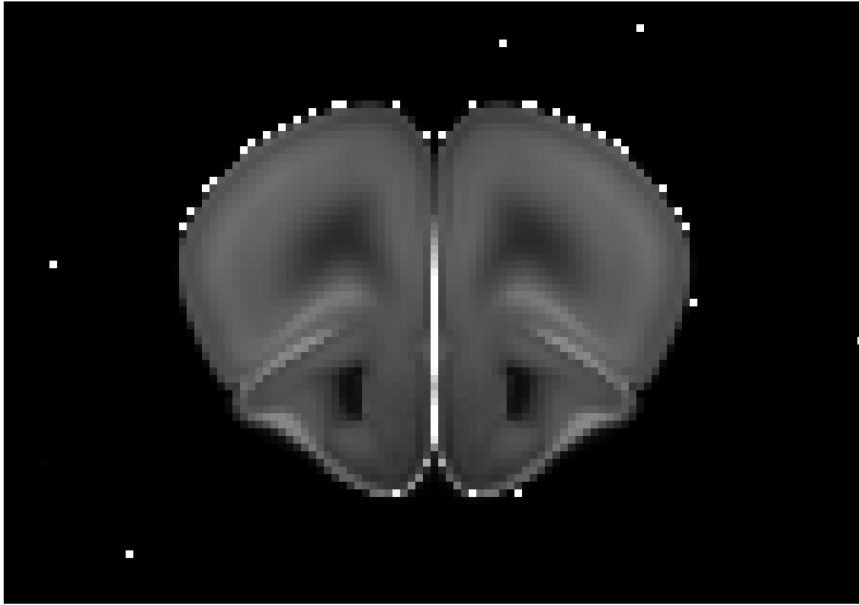
However, some of the defaults are different. Notably, the default mask is the non-zero entries in the matrix. This improves the plotting of the annotation data:

```
figure; plotCcfAnnotation(getAllenTemplate('annotation'));
```



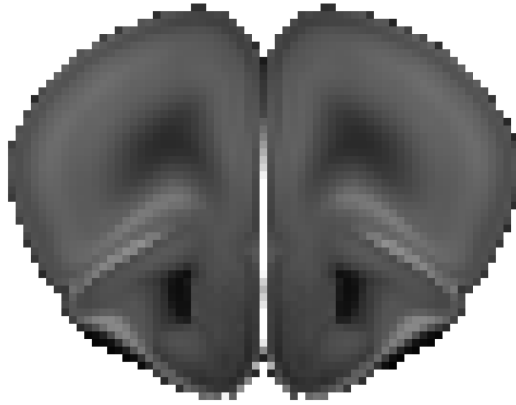
It worsens the plotting of the `average_template` data:

```
figure; plotCcfAnnotation(getAllenTemplate('average_template')); colormap(gray);
```

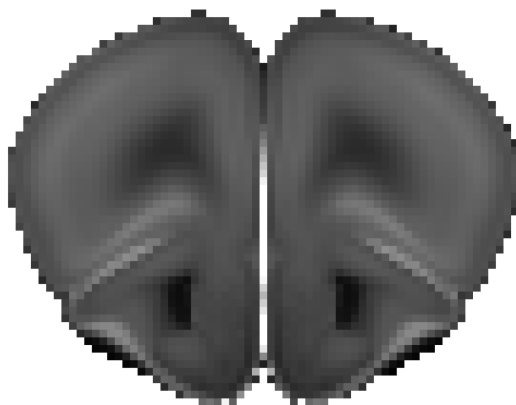


But this can be readily fixed by using the annotation data as a mask for the `average_template`:

```
figure; plotCcfAnnotation(getAllenTemplate('average_template'), 'mask',  
getAllenTemplate()); colormap(gray);
```

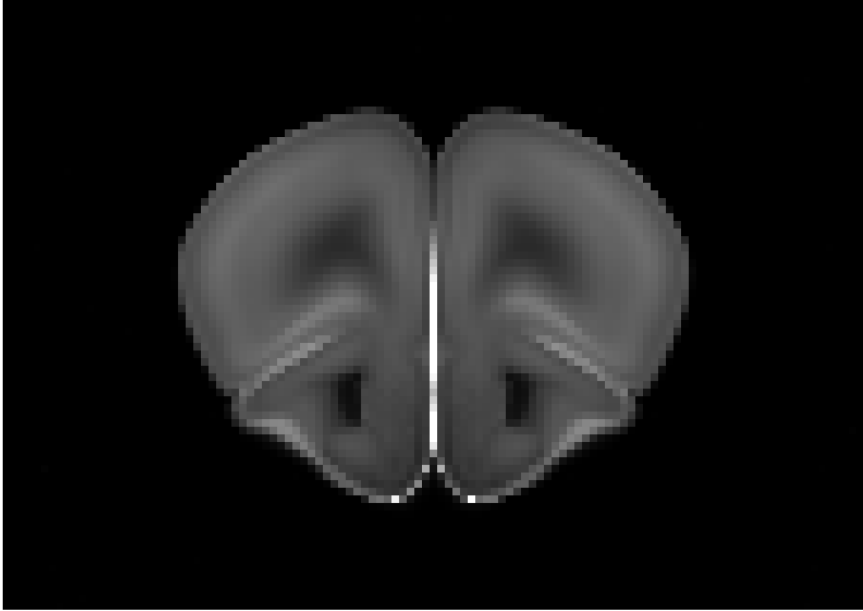



```
figure; plotCcfAnnotation(getAllenTemplate('average_template'), 'mask',  
getAllenTemplate('annotation')); colormap(gray);
```

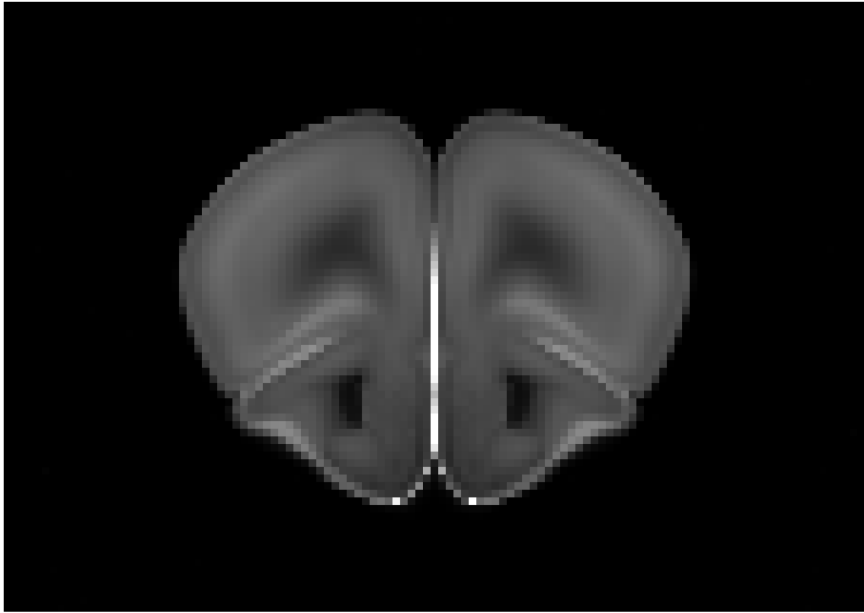


or by plotting the whole volume:

```
figure; plotCcfAnnotation(getAllenTemplate('average_template'), 'mask', @(x)  
true(size(x))); colormap(gray);
```

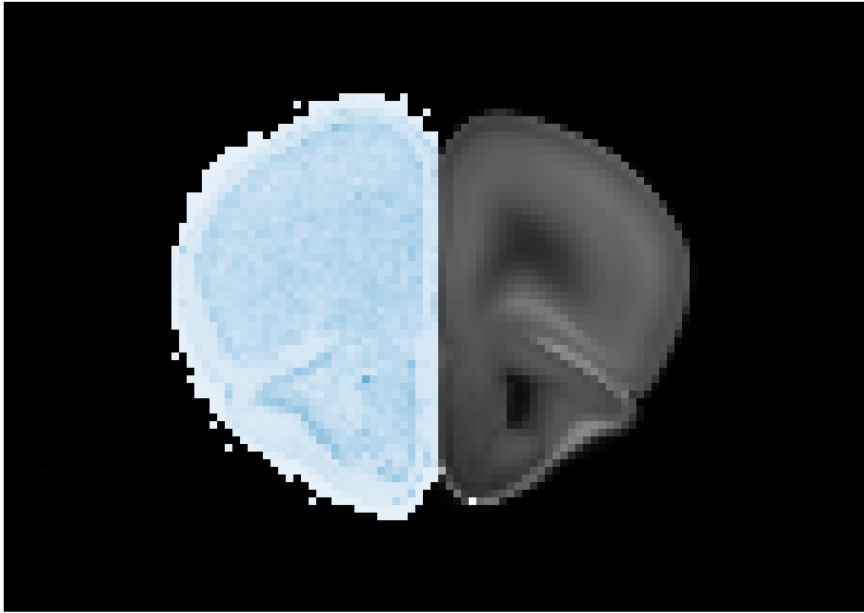


```
figure; plotCcfAnnotation(getAllenTemplate('average_template'), 'mask',  
ones(size(getAllenTemplate()))); colormap(gray);
```



Plotting both background and foreground data can be done by using the two functions together. `plotAllenSlice` is recommended for the background, as it is in true colour. `plotCcfAnnotation` is recommended for the foreground, as it is scaled colour (and so can be modified post-runtime with `colormap`, `clims` etc).

```
V = getAllenTemplate('ara_nissl', 100, 2);  
figure;  
plotAllenSlice();  
hold on;  
plotCcfAnnotation(V);  
colormap(sky);
```



```
% % A large list of foreground and background plots/options
% figure; plotCcfAnnotation(getAllenTemplate('annotation'));
% figure; plotCcfAnnotation(getAllenTemplate('annotation'), 'mask',
getAllenTemplate('annotation'));
% figure; plotCcfAnnotation(getAllenTemplate('annotation'), 'mask', @logical);
% figure; plotCcfAnnotation(getAllenTemplate('annotation'), 'mask', @(x) true(size(x)));
%
% figure; plotCcfAnnotation(getAllenTemplate('average_template')); colormap(gray);
% figure; plotCcfAnnotation(getAllenTemplate('average_template'), 'mask',
getAllenTemplate('annotation')); colormap(gray);
% figure; plotCcfAnnotation(getAllenTemplate('average_template'), 'mask', @logical);
colormap(gray);
% figure; plotCcfAnnotation(getAllenTemplate('average_template'), 'mask', @(x)
true(size(x))); colormap(gray);
%
% figure; plotAllenSlice('annotation', 'cmap', parula);
% figure; plotAllenSlice('annotation', 'cmap', parula, 'mask',
getAllenTemplate('annotation'));
% figure; plotAllenSlice('annotation', 'cmap', parula, 'mask', @logical);
% figure; plotAllenSlice('annotation', 'cmap', parula, 'mask', @(x) true(size(x)));
%
% figure; plotAllenSlice('average_template');
% figure; plotAllenSlice('average_template', 'mask', getAllenTemplate('annotation')); % bad
% figure; plotAllenSlice('average_template', 'mask', @logical);
```

```
% figure; plotAllenSlice('average_template', 'mask', @(x) true(size(x)));
```

Creating animations using `videofig(s)` from `nihelp`

These animations cannot be created in a live script, but have a look at the examples below and in the test files.

```
% videofig(132, @(n) plotAllenSlice('average_template', 'slice', n));

%% Compare high res nissl with low res MRI
% V = getAllenTemplate('ara_nissl',50).*getAllenRMask(50); % example volumetric data
% videofigs(132, @(n) cla, @(n) plotAllenSlice('dim', 'c', 'slice', n, 'resolution',
100), ...
%     @(n) hold('on'), @(n) plotCcfAnnotation(V, 'dim', 'c', 'slice', n*2-1, 'resolution',
50), ...
%     @(n) colormap('sky'));
```